

What Did Each Simulation Estimate?

Unit 2 · Topic 2.3 · THE PROBLEM

Suppose each of five students can be treated as independently selected, and each has probability 0.12 of being left-handed. Two students use 1,000 simulation trials to estimate the probability that at least 2 of the 5 students are left-handed. Their plans and results are shown.

Results: Student 1 = $87/1,000$; Student 2 = $94/1,000$

Plan	L assignment	Trial; success
1	0 of 0–9	5 digits; 2+ L
2	00–11 of 00–99	5 pairs; exactly 2 L

FIRST TAKE — one sentence before you work the parts

Which simulation result would you trust more for the question asked—Student 1, Student 2, or neither? Name one reason.

- DOK 1** (a) List the left-handed counts included in at least 2 of 5. State the chance one student is left-handed.
- DOK 2** (b) Find each reported estimate. For each plan, identify either a mismatch in the chance assignment or a mismatch in the event counted.
- DOK 3** (c) Would you trust either plan for the question asked? Explain both mismatches and why the larger reported estimate does not validate a plan. Describe one corrected trial, including the random labels and the event counted. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.